

Takes Two to Tango: Predicting Player Chemistry in Soccer

from Event Data

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September 2026

Introduction

Can the chemistry between two soccer players be predicted before they play together? Forecasting player compatibility can improve scouting and squad building by helping clubs assess not only a player’s individual quality, but also how effectively they are likely to combine with prospective teammates. Existing work predicts chemistry from demographic variables such as nationality, language or age. We propose a data-driven framework that predicts chemistry from past on-pitch actions, developed as a scouting tool for Real Racing Club de Santander.

Methods

We use Opta event data from six European leagues between 2021/22 and 2025/26, covering 6,167 players and 153 teams. Realized offensive chemistry is measured using Joint Offensive Impact per 90 minutes (JOI90), a metric based on the VAEP framework that captures the change in scoring probability generated by consecutive actions involving both players. Our sample comprises 987 pairs who had not played together in the preceding season and subsequently accumulated at least 1,500 shared minutes.

Five compatibility features are constructed from prior-season data: action complementarity (whether one player tends to pass to areas where the other frequently receives the ball), positional proximity, role-based affinity (how often each player passes to the other’s role), forward-progression alignment (whether one player’s passes travel as far forward as the passes the other receives), and team-style similarity. A two-stage linear model first regresses JOI90 on Transfermarkt market values to adjust for individual player quality, then models the residual on the compatibility features.

Results

The model exhibits strong out-of-sample predictive performance. It achieves an RMSE of 0.03715, a 20.8% reduction against a naïve baseline that predicts the training-set mean, and explains 37.3% of the out-of-sample variation in JOI90. The model also beats a market-values-only model (Table 1), as well as the results reported in related works. Results hold across alternative specifications and estimators.

Spearman correlation between predicted and realized chemistry is 0.60 ($p < 0.001$). Of the ten pairs with the highest predicted chemistry in the test set, five are also in the top ten pairs with the highest actual chemistry (Table 2).

Model	RMSE	Improvement over naïve baseline
Naïve baseline	0.04692	–
Market values only	0.04200	10.5%
Full model	0.03715	20.8%

Table 1: Model predictive performance

Predicted rank	Pair	Season	Realized rank
1	Jude Bellingham & Kylian Mbappé	24/25	#12
2	Harry Kane & Jamal Musiala	23/24	#1
3	Lautaro Martínez & Marcus Thuram	23/24	#7
4	Alexis Mac Allister & Dominik Szoboszlai	23/24	#37
5	Dominik Szoboszlai & Luis Díaz	23/24	#26
6	Darwin Núñez & Dominik Szoboszlai	23/24	#6
7	Cole Palmer & Jadon Sancho	24/25	#13
8	Kylian Mbappé & Lucas Hernandez	23/24	#15
9	Jude Bellingham & Toni Kroos	23/24	#2
10	Marcel Sabitzer & Niclas Füllkrug	23/24	#3

Table 2: Player pairs with highest predicted chemistry in the test set

Conclusion

Players’ on-pitch behaviors contain meaningful information about their likely compatibility with teammates they have not yet played alongside. The proposed framework thus provides clubs with a practical tool for comparing transfer targets and identifying players who fit their squads.